

2025 AMC 10B Problem 9 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10B Problem 9. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10B Problem 9

Problem:

How many ordered triples of integers (x, y, z) satisfy the following system of inequalities?

$$-x - y - z \leq -2$$

$$-x + y + z \leq 2$$

$$x - y + z \leq 2$$

$$x + y - z \leq 2$$

Answer Choices:

- A. 4
- B. 8
- C. 11
- D. 15
- E. 17

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Solution:

First, we rewrite the inequalities to obtain the following:

$$-x - y - z \leq -2, \quad (1)$$

$$-x + y + z \leq 2, \quad (2)$$

$$x - y + z \leq 2, \quad (3)$$

$$x + y - z \leq 2. \quad (4)$$

We can start by bounding our variables. By adding (2) and (3), we get that

$$2z = (-x + y + z) + (x - y + z) \leq 2 + 2 = 4 \quad \implies \quad z \leq 2$$

By symmetry (adding (2), (4) and (3), (4)), we get that $x \leq 2$ and $y \leq 2$.

By adding (2) and (1), we get that

$$-2x = (-x + y + z) + (-x - y - z) \leq 2 - 2 = 0 \quad \implies \quad x \leq 0$$

By symmetry (adding (1), (3) and (1), (4)), we get that $y \geq 0$ and $z \geq 0$.

Thus

$$0 \leq x, y, z \leq 2 \quad \implies \quad x, y, z \in \{0, 1, 2\}.$$

We must still have

$$x + y + z \geq 2.$$

We can note that since our inequalities are symmetric in x, y, z (swapping them won't do much), we can take cases on $\min(x, y, z)$.

Case 1. $\min(x, y, z) = 0$. In this case, without loss of generality (since we're dealing with symmetry), suppose $x = 0$. Then, our inequalities become

$$-y - z \leq -2 \quad y + z \leq 2 \quad -y + z \leq 2 \quad y - z \leq 2$$

Note that since $y, z \in \{0, 1, 2\}$, the last two are automatically satisfied. The first two combine to say that $y + z = 2$ (the first becomes $y + z \geq 2$), so we have 2 distinct (up to permutation) possibilities: $(0, 0, 2), (0, 1, 1)$. Each of these can be re-ordered in 3 ways, for a total of 6 solutions here.

Case 2. $\min(x, y, z) = 1$. Without loss of generality suppose $x = 1$. In this case, our inequalities become

$$y + z \geq 1 \quad y + z \leq 3 \quad -y + z \leq 1 \quad y - z \leq 1$$

Note that the first inequality is automatically satisfied as $y, z \geq 1$, and the second inequality is satisfied if not both y and z are 2. We note that both possibilities $(1, 1, 1)$ and $(1, 2, 1)$ are solutions, for a total of 4 solutions in this case.

Case 3. $\min(x, y, z) = 2$. In this case $x = y = z = 2$, and we can verify this works, for an additional solution.

This gives a total of

$$6 + 4 + 1 = \boxed{(C) \ 11}$$

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